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Evaluation of Ozone Therapy in Endodontic Treatment of Teeth with Necrotic Pulp and Apical Periodontitis: A Randomized Clinical Trial

SIGNIFICANCE

Aqueous ozone has the potential to be used as an antimicrobial and anti-inflammatory agent with correct application time and activation technique after traditional cleaning, shaping, and irrigation in postendodontic main management, especially in patients undergoing single-visit endodontic treatment.

KEY WORDS

Manual dynamic activation; ozone treatment; passive ultrasonic activation; randomized controlled trial; sonic activation; visual analog pain scale

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Pain is an undesirable and unfortunate yet common sensation after root canal treatment. It may commence after a few hours to days after treatment and is at all times an unpleasant experience, both for the patient and the clinician^{1,2}. The frequency of postoperative pain in patients can range from mild discomfort (3%–58%)³ to severe pain (up to 6%)⁴. This frequency is still higher in nonvital teeth (40%) in comparison with vital teeth (9%)⁵.

Pain is most likely due to the activation of chemical mediators like prostaglandins released from damaged tissues, possibly due to endodontic instrumentation, periapical contamination by caustic irrigants, and intracanal medications. These inflammatory mediators can activate and sensitize nociceptors, resulting in peripheral sensitization leading to pain. Bacteriologic factors play a critical role in increasing the intensity of periapical inflammatory reactions⁶. Lipopolysaccharide released by gram-negative bacteria is a major virulence factor exerting a plethora of biological effects amplifying the inflammation^{7,8}.

To improve the antimicrobial capacities and facilitate faster periapical healing, innovative approaches in irrigants and irrigation techniques have been proposed. Ozone is one such irrigant that is an emerging and promising disinfectant in endodontics, mainly because of its powerful bactericidal action and low cytotoxicity. At higher concentrations, its bactericidal action is much faster, and it is 1000 times

more powerful than most disinfectants. A single ozone molecule is equivalent to 3000–10,000 chlorine molecules, thereby acting around 3500 times faster on microorganisms. It not only damages the bacterial cell membranes by ozonolysis but also causes oxidation of intracellular proteins, leading to organelle function loss. It has selective action against microbial cells and hence does not affect human cells, resulting in minimal cytotoxicity and high biocompatibility with oral tissues.

Furthermore, it has high efficiency against antibiotic-resistant strains, and its effect increases in acidic pH. Its other advantages include improvement in cell immunity and humeral systems, the proliferation of immune-competent cells, immunoglobulin synthesis, activation of the macrophage, and a lack of mutagenicity. However, some of the drawbacks of ozone are its unstable nature in an aqueous state, inconsistent activity in biofilms and gram-positive bacteria, no residual effect, poor diffusivity to deeper areas of the dentinal tubules, and rapid deterioration in the presence of organic tissue. The current literature suggests that ozone may not be an alternative to contemporary irrigants, but it has the potential to be a complementary disinfectant⁹.

Conventional irrigation regimens fail to ensure complete canal disinfection due to limitations like air bubble formation and vapor locks, which prevent the movement of fluid into the narrow confines of fins, webs, isthmuses, and lateral canals¹⁰. Thorough movement of the irrigation solution with physical agitation enhances its chemical and biological action by increasing the contact between the fluid and the root canal walls and raising the temperature of the irrigant, accelerating its overall action¹¹.

There is a need to explore the possible role of ozone in periapical healing and pain management because of its high oxidative power, which increases adenosine triphosphate synthesis, resulting in improvement of the cell metabolism and accelerating the repair process by angiogenesis stimulation. The efficiency of ozone as an antimicrobial agent is well researched; however, the literature is lacking regarding its use in postendodontic pain management. Numerous trials favor its use as a disinfection agent;^{12,13} however, some demonstrate unfavorable outcomes also^{14,15}. Evidence in regard to its effect on postoperative pain control when used with activation in patients undergoing single-visit endodontics (SVE) is scarce. Hence, this study aimed to compare postoperative pain in

patients undergoing SVE with necrotic pulp and apical periodontitis after the application of aqueous ozone activated by manual, sonic, and ultrasonic activation versus no ozone treatment.

The primary objective of this study was to evaluate and compare postendodontic pain in patients undergoing SVE after aqueous ozone application (no agitation, manual, sonic, and ultrasonic) versus no ozone treatment at baseline; 6, 12, 24, and 48 hours; and 7 days. The secondary objective was to analyze the effect of other variables like sex (male/female), age group, and arch (maxilla/mandible) on postendodontic pain.

MATERIALS AND METHODS

Selection of Subjects

The study was conducted in the Department of Conservative Dentistry and Endodontics, Pacific Dental College and Hospital, Udaipur, Rajasthan, India, from January 2021 to August 2021. Before starting the study, approval of the research protocol and ethical clearance (PDCH/19/EC-140) were obtained from the institutional review board. The study was registered in [clinical.trials.gov](https://clinicaltrials.gov) (REF/2021/03/041572) and followed Consolidated Standards of Reporting Trials 2010 guidelines.

All trial participants were informed about the purpose of the study, the treatment protocol to be followed, the benefits and risks associated, and alternative treatment options. Willing subjects of this blinded, parallel, randomized controlled trial (allocation ratio 4:1) were requested to sign an informed consent form before the study.

Sample Size Estimation

To determine the sample size, a pilot study with 40 patients was conducted with 5 groups (ozone treatment with no activation [NA], ozone treatment with manual dynamic activation [MDA], ozone treatment with passive ultrasonic activation [PSA], ozone treatment with sonic activation [SA], and no ozone treatment [control]) with 8 subjects each. The effect size F was estimated as 0.383 after calculating the partial eta square for group * time interaction having the value of 0.128. The sample size was estimated by using G Power (version 3.19.7) software. The sample size of 110 was calculated at a 95% confidence interval with a power of 90% having an effect size of $F_4 = 0.383$. Anticipating dropouts during the follow-up, the sample size was increased by 10%, making it 121. Hence, a final sample size of 24 subjects per group was estimated for the study.

Inclusion and Exclusion Criteria

Patients in the age range of 18–65 years with good general health available for follow-up and willingness to give consent for the study were selected. Single-rooted teeth with necrotic pulp and apical periodontitis, periapical index score <3 , sound periodontal apparatus, and sufficient coronal tooth structure were included.

Patients with systemic diseases, immunocompromised conditions, pregnancy, a history of allergic reactions to the drugs/irrigants, a sinus tract, an acute periapical abscess, and teeth with calcified canals/immature apex/mobility were not included in the study. Patients with recent myocardial infarction, acute alcohol intoxication, active bleeding, uncontrolled hyperthyroidism, thrombocytopenia, or glucose 6 phosphatase deficiency (contraindications of ozone therapy) were excluded.

Subject Allocation, Blinding, and Randomization Method

A secondary investigator not involved in the trial evaluated the eligibility of the patients based on history and clinical and radiographic examination. Pulpal diagnosis of each subject was determined using cold and electric pulp testing. Moreover, the absence of bleeding from the root canals during access cavity preparation was also considered as a confirmatory clinical diagnosis. For radiographic examination, intraoral periapical radiographs were taken, and areas other than the periapex were covered with black photographic paper. Using reference photographs, the periapical region was scored between 1 and 5 categories according to the periapical index¹⁶.

To ensure randomization and allocation concealment, patients randomly picked a computer-generated, randomly sequenced, sealed opaque envelope, which contained the group code. This envelope was given to the primary investigator after completion of shaping and cleaning just before ozone application. The patients were blinded and were unaware of the specific treatment protocol they would be receiving depending on their group allotment. The patients of the control group were irrigated with saline similar to the no agitation group as a placebo treatment to ensure blinding.

The visual analog scale (VAS) scale was used to assess pain at baseline; 6, 12, 24, and 48 hours; and 7 days. All the participants were instructed to put a mark on the VAS scale corresponding to pain intensity. On completion of the study, a blinded assessor filled the master chart with the VAS scores of each

patient without knowing the actual group of the subject. The statistician concerned with the evaluation of pain intensity was blinded as well.

Treatment Protocol

Of the 177 patients assessed for eligibility for the study, 34 patients did not meet the inclusion criteria, and 23 patients declined to participate in the trial; the remaining 120 patients were randomly divided into 5 groups. The primary investigator performed all the treatments in a single visit. Under rubber dam isolation, a standard access cavity was prepared after achieving adequate anesthesia with 2% lidocaine with 1:100,000 epinephrine (Lignox; Indoco, Warren, Mumbai, India). An electronic apex locator (Root ZX mini; J Morita, Tokyo, Japan) was used for working length estimation followed by radiographic confirmation. Maintaining the apical patency using a #10 K-file, cleaning and shaping were completed with a crown-down technique using rotary nickel-titanium files (ProTaper; Dentsply Maillefer, Ballaigues, Switzerland) and 17% EDTA following the manufacturer's instructions. With each step of instrumentation, 2 mL 5.25% sodium hypochlorite (NaOCl) was used for irrigation with a 27-G side-vented needle (Dentsply, Tulsa, OK). Apical preparation was completed to 3 sizes greater than the first binding apical file. After completion of instrumentation, patients were assigned to 1 of the following groups based on randomized group allotment:

- Group 1: NA
- Group 2: MDA
- Group 3: PUA
- Group 4: SA
- Group 5: the control group

Aqueous ozone was obtained by bubbling ozone in the gaseous state (4 mg/mL) in distilled water for 15 minutes using an ozone generator (Dentozone, Analytical & Medical Technology, Mumbai, Maharashtra, India). Individual groups were irrigated with 15 mL ozonated water for 3–5 minutes in each of the canals, keeping the tip 2 mm short of the working length.

MDA was performed with well-fitting gutta-percha (master apical cone) by short push-pull strokes at an approximate rate of 100 strokes per minute. PUA was achieved with a stainless steel ultrasonic file ISO size #10 (Satelec Acteon Group, Merignac Cedex, France) after 3 cycles of 20 seconds each at a power setting of 3. SA was performed using the Pro Agitator Tip System (Innovations Endo Ltd, Nasik, Maharashtra, India).

Root canals were dried with sterile paper points, coated with AH Plus (Dentsply

Maillefer) sealer, and obturated at the same appointment using lateral condensation. Teeth were reduced from occlusion and temporized with Cavit (3M ESPE, St Paul, MN).

No postoperative analgesics were advised to the patients. A reserve medication (ibuprofen) and a backup phone number were given in case of severe pain or an emergency. The patients were excluded from the study if the reserve medication was taken within 7 days after treatment.

Postendodontic Pain Assessment

Pain assessment was performed using the VAS scale, which consisted of a 10-cm line anchored by 2 extremes, “no pain” in the beginning and “pain as bad as it could be” in the end. Based on the level of pain perceived, the patient marked a position on the line at specific intervals (ie, before treatment [baseline]; at 6, 12, 24, and 48 hours; and 7 days after treatment). Patients were recalled for follow-up after 2 days and 1 week following treatment to return the pain diary and for a clinical evaluation.

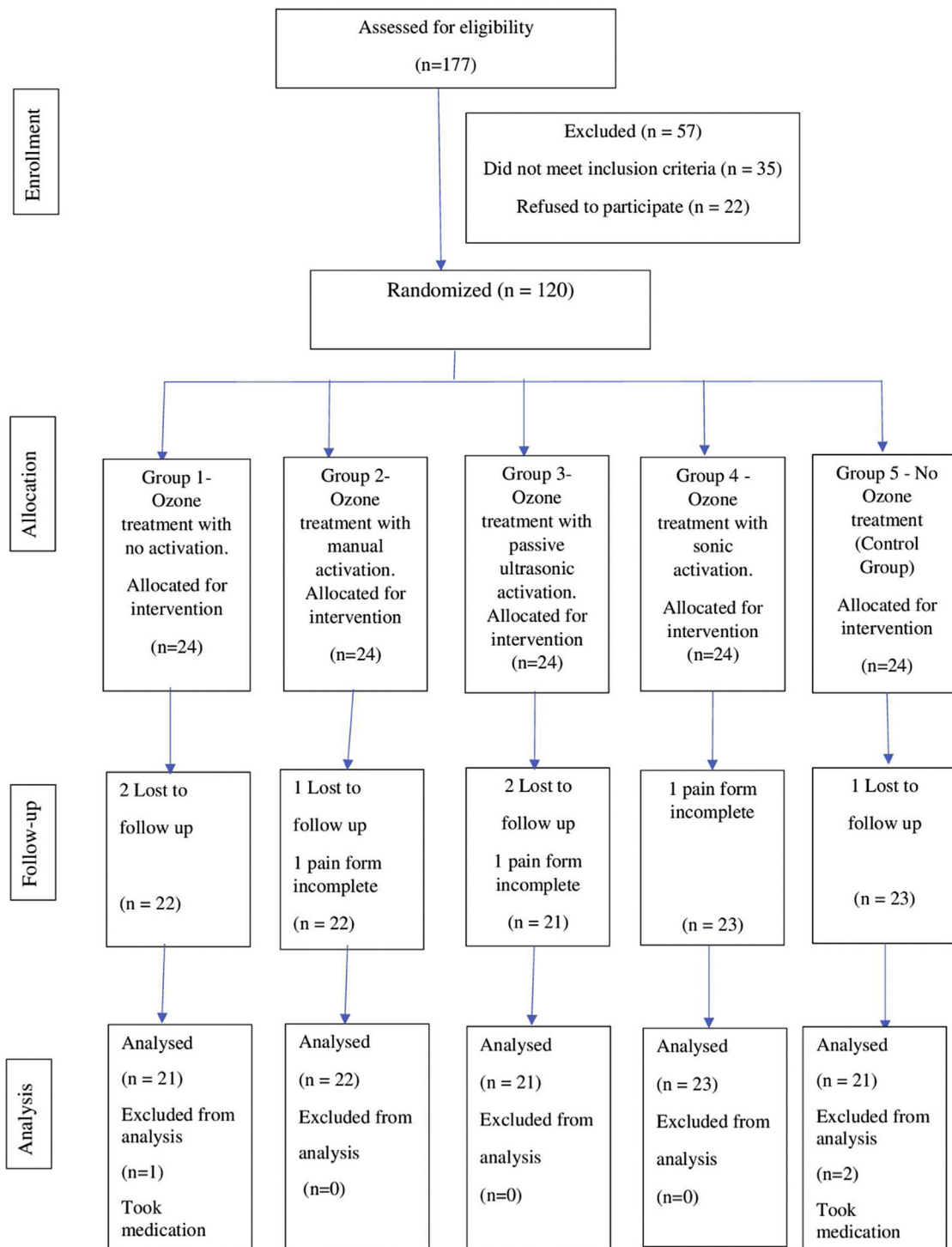


FIGURE 1 – The Consolidated Standards of Reporting Trials flow diagram.

Data presented as mean ± standard deviation.

CONCLUSION

Within the scope of this study, it can be concluded that the application of activated ozone resulted in less postendodontic pain in teeth with necrotic pulp and apical periodontitis than no ozone treatment. The study outcome suggests that aqueous ozone applied with an ultrasonic/sonic technique can be recommended as an additional application protocol for better pain control, particularly in SVE. Its simple application procedure, with a suitable irrigation time and a relative absence of discomfort, increases patient acceptability and compliance. Moreover, its noninvasive nature and good biocompatibility make it an ideal treatment alternative for clinicians while treating blunderbuss canals, internal and external resorption cases, perforations, extraoral, intraoral sinuses, and primary teeth.

Limitations of the study include unequal age and sex distribution of patients in the individual groups, which could contribute to different pain thresholds resulting in variation in outcomes. Because the study results were based on subjective scores recorded by the patients, participant bias was a possible drawback. Another potential bias was operator bias; in this interventional study, the operator could not be blinded for the procedures being performed. Future clinical trials examining varied clinical scenarios with well-defined parameters and with minimal bias should be conducted to validate the potential role of ozone therapy in pain control, particularly in postendodontic treatment.

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The authors deny any conflicts of interest related to this study.

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